

PART 2 — INTERNAL STUDY REVIEW MEMO

To: Scholera Product & Research Team
From: AI/ML Research Intern
Subject: Review of the Pre-Class Primer Trial
Date: August 2026

EXECUTIVE SUMMARY

The four-week pre-class primer trial provides a promising signal, but it does not establish that primers caused the improvement in student outcomes.

Sections receiving primers had a 7.4 percentage-point higher on-time assignment submission rate than control sections. Students who voluntarily rated the primer also gave it an average usefulness score of 4.1/5.

These results are encouraging enough to justify further experimentation. However, they are not strong enough to support the proposed graduate-specific rollout, tuning, or pricing decision.

The graduate subgroup is particularly uncertain. The reported increase from 74.1% to 95.8% looks substantial, but it comes from only 9 students in the control group and 4 in the treatment group. This is too small a sample to confidently conclude that graduate students benefit more from primers.

My recommendation is to continue testing the primer feature, but run a larger and better-controlled experiment before making graduate-specific product or pricing decisions.

1. WHAT THE STUDY SUPPORTS

The study provides two useful signals.

First, treatment sections had a 7.4 percentage-point higher on-time assignment submission rate than control sections. The trial covered 11 sections over four weeks, with six treatment sections and five control sections.

Second, students who chose to provide a usefulness rating generally viewed the primer positively, with an average rating of 4.1 out of 5.

The undergraduate breakdown also points in the same direction. For first- and second-year students, on-time submission increased from 71.2% in control sections to 78.9% in treatment sections. For third- and fourth-year students, it increased from 79.5% to 85.1%.

Taken together, these results make the primer feature worth investigating further. There is enough signal here that I would not recommend abandoning the feature.

However, I would describe the result as an association rather than a causal effect. The safest conclusion is:

"The trial found an encouraging association between receiving pre-class primers and higher on-time assignment submission."

I would not state that the primers "increased" on-time submission by 7.4 percentage points, because the study design does not establish causality.

2. WHAT THE STUDY DOES NOT SUPPORT

The main limitation is that treatment and control sections were not isolated from other changes occurring during the trial.

Two treatment sections also introduced weekly quizzes. If submission behavior improved in those sections, some or all of the improvement could have resulted from the quizzes rather than the primers.

There was also a control section whose professor was on leave for two of the four weeks. This creates another difference between treatment and control conditions that is unrelated to the primer.

The study also reports that sections where professors posted primers within 24 hours of class performed better. However, timing was not controlled. More engaged professors may also be more likely to post earlier and may influence student engagement in other ways. Therefore, the observed relationship between timing and outcomes cannot be interpreted as evidence that earlier posting itself caused the improvement.

Another important limitation is that the study tracked whether primers were sent, but not whether students actually opened or read them. Therefore, the study cannot tell us whether the primer content itself affected behavior or whether simply receiving the primer was associated with the outcome.

Finally, usefulness ratings were voluntary. Approximately two-thirds of treatment students provided a rating. The 4.1/5 result is therefore useful as a signal of student sentiment, but it should not automatically be treated as representative of every student who received a primer.

3. THE GRADUATE RESULT NEEDS PARTICULAR CAUTION

The internal write-up highlights the graduate cohort as the strongest result. On-time submission increased from 74.1% in control to 95.8% in treatment, and usefulness was rated 4.9/5.

The problem is the sample size.

There were only:

- 9 graduate students in the control group
- 4 graduate students in the treatment group
- 4 graduate students providing usefulness ratings

With numbers this small, a few individual students can have a large effect on the observed percentages.

Therefore, I would treat the graduate result as a hypothesis-generating observation rather than evidence that primers are especially effective for graduate students.

This distinction matters because the proposed next step goes beyond simply continuing the experiment. The internal write-up recommends prioritizing graduate sections, building graduate-specific tuning, and potentially charging more for the graduate tier.

That recommendation is substantially stronger than what the current data can support.

The graduate result is interesting enough to investigate, but not strong enough to make it the basis for a product or pricing decision.

4. WHAT I WOULD DO NEXT

I would continue testing the primer feature, but redesign the experiment to address the uncertainties in the current study.

First, increase the number of participating sections. A larger sample would make the results less sensitive to unusual professors, classes, or individual students.

Second, randomize treatment assignment where feasible. Random assignment would make it easier to distinguish the effect of the primer from pre-existing differences between treatment and control groups.

Third, standardize primer timing. The current study suggests that posting within 24 hours may be associated with better outcomes, but timing was not controlled. The next study should either standardize timing or deliberately test different timing windows.

Fourth, track actual engagement. Record whether students opened the primer and, if possible, how much they interacted with it. This would allow us to distinguish between:

Primer sent -> outcome

and

Primer actually consumed -> outcome.

That distinction is important because Scholera ultimately wants to understand what part of the product creates value, not just whether an intervention was assigned.

Fifth, control concurrent interventions. Weekly quizzes or other changes to course structure should either be kept consistent across treatment and control sections or explicitly recorded and accounted for.

Sixth, pre-specify subgroup analysis. If Scholera wants to know whether primers work differently for undergraduate and graduate students, that question should be defined before looking at the results and tested with a sufficiently large sample.

The current graduate result should therefore be treated as a reason to investigate that hypothesis, not as confirmation of it.

5. ADDITIONAL OBSERVATIONS

There are two broader points worth carrying into the next experiment.

First, the current study measures the effect of being sent a primer, not necessarily the effect of consuming one. This means the experiment does not yet tell us much about the mechanism behind the observed improvement. If many students receive a primer but never open it, a positive treatment effect would need a different explanation than if students who actively read the primer are the ones who change their behavior.

Second, the study's strongest business recommendation is based on its smallest subgroup. The graduate result is the most dramatic number in the report, but it also comes from the least informative sample. This is exactly the kind of situation where a large percentage difference can look more convincing than the underlying evidence deserves.

A useful next experiment should therefore be designed not just to produce a larger headline number, but to answer the specific uncertainties left by this trial.

6. BOTTOM LINE

The study gives Scholera a good reason to keep testing pre-class primers, but not a good reason to make the proposed graduate-specific product or pricing decision yet.

The most defensible conclusion is:

"Primers are promising, but their causal impact has not been established."

The 7.4 percentage-point difference in on-time submission and the positive usefulness ratings justify another experiment. However, the uncontrolled differences between sections, voluntary usefulness ratings, lack of actual engagement tracking, and especially the very small graduate sample limit how strongly we can interpret the results.

I would therefore continue the primer experiment, improve its measurement and experimental controls, and defer graduate-specific rollout, tuning, and pricing decisions until a larger study provides stronger evidence.

The next study should answer three questions clearly:

1. Does the primer itself cause an improvement in student behavior?
2. Do students actually engage with the primer, and does engagement explain the outcome?
3. Does the effect genuinely differ between undergraduate and graduate students?

Until those questions are answered with stronger evidence, the current study should be treated as a promising signal rather than a basis for a major product decision.